C Programming

Ch.6 Conditional Statements

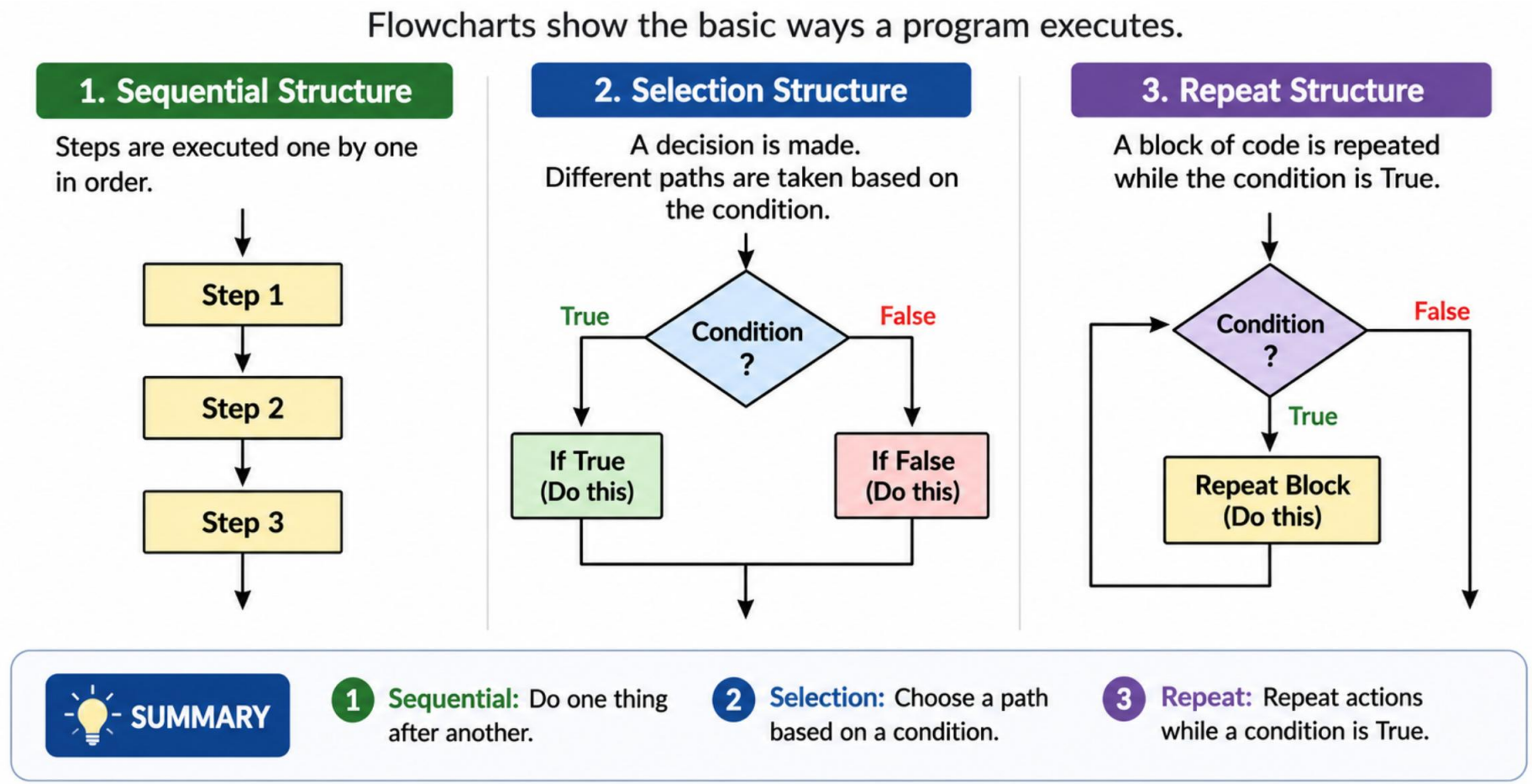
Operators | if / switch | while, do-while, for

Ch. 6

Conditional Statements

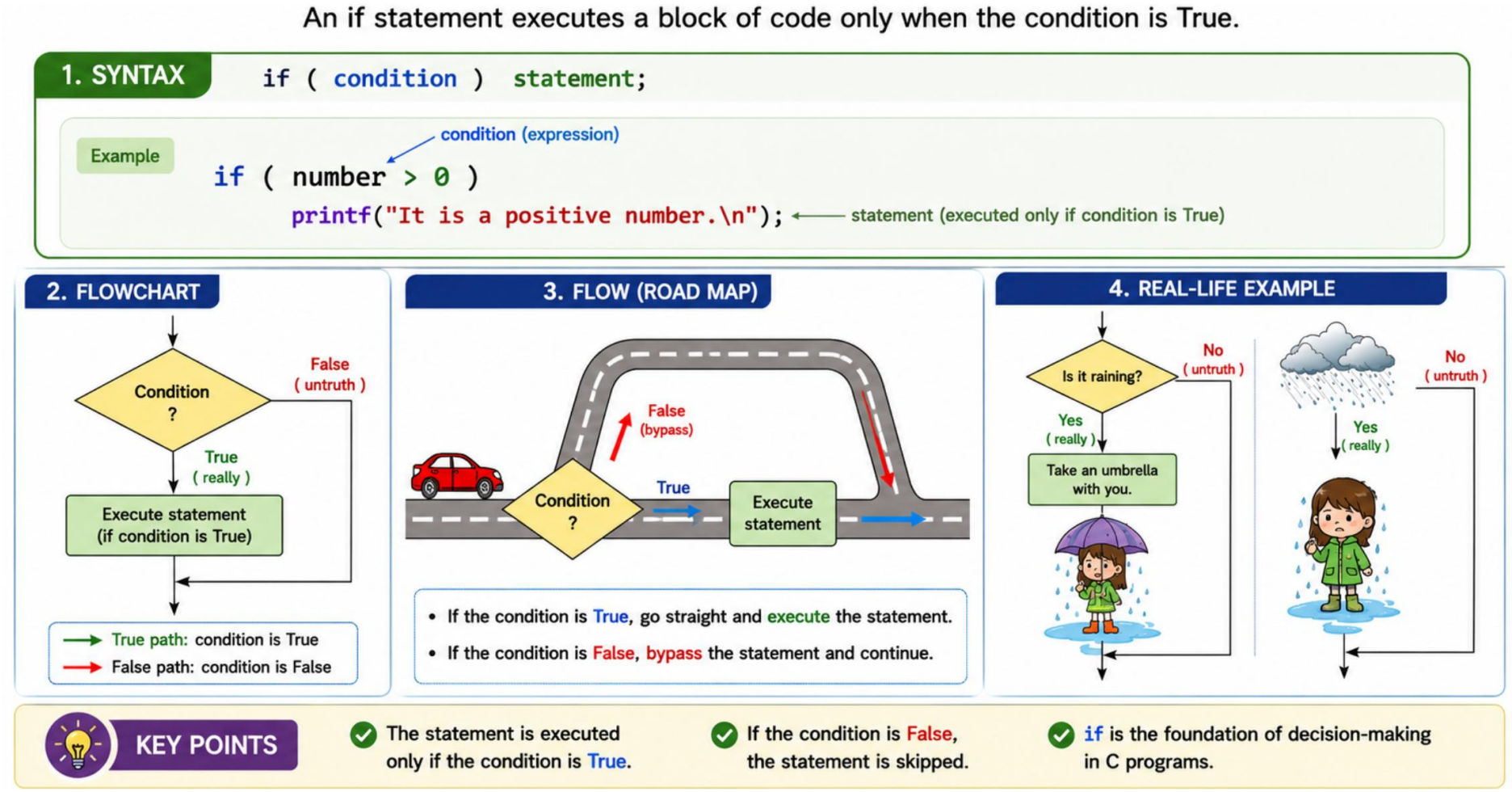
Conditional Statement

- If a program does not have a selection structure, the program will always repeat the same actions.
- Three types of control structures: sequential structure, selection structure, repeat structure.



Conditional Statement: if statement

- **if statement**: in everyday life, there are many instances where decisions must be made based on conditions.



if Statement Example

```c
#include < stdio.h >
int main( void )
{
    int number;
    printf ( "Enter an integer:" );
    scanf ( "%d" , &number);

    if ( number > 0 ) {
        printf ( "It is a positive number." );
    }
    printf ( "The entered value is %d.", number );

    return 0;
}
```

if the if-statement ends, the following line is always executed.

Compound Statement (Block)

- Compound statement: grouping statements using curly braces { }, also called a block.
- If the condition is true, all statements inside the block are executed:
-

```
if (score >= 60) { printf("You passed.\n"); printf("You can also get a scholarship.\n");  }
```

```
if (score >= 60) {
    printf("You passed.\n");
    printf("You can also get a scholarship.\n");
  }
```

Conditional Statement: if-else Statement

An if-else statement executes one block when the condition is True, otherwise another block.

1. SYNTAX

```
if ( condition ) statement1; else statement2;
```

Example

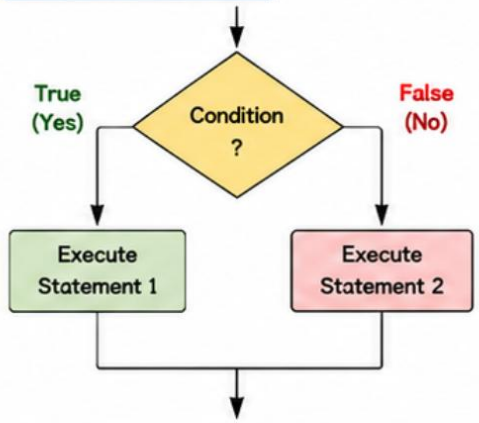
```
if ( number > 0 )
    printf("It is a positive number.\n");  // statement1 (executed if condition is True)
else
    printf("Not a positive number.\n");    // statement2 (executed if condition is False)
```

condition (expression)

If the condition is True, statement1 is executed.

If the condition is False, statement2 is executed.

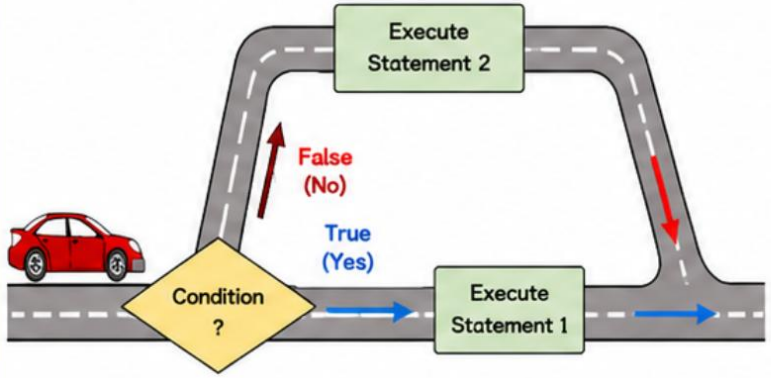
2. FLOWCHART



True path: condition is True

False path: condition is False

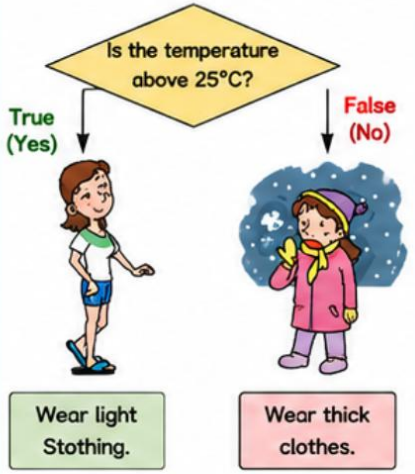
3. FLOW (ROAD MAP)



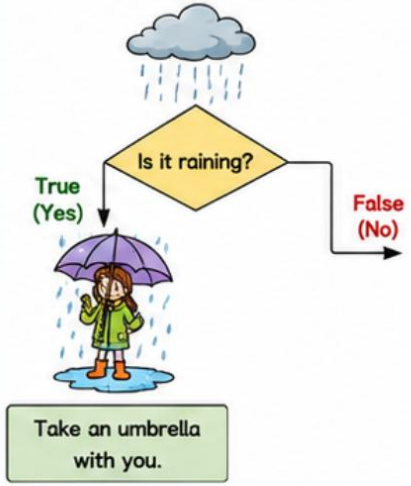
- If the condition is True, follow the blue path and execute Statement 1.
- If the condition is False, follow the red path and execute Statement 2.

4. REAL-LIFE EXAMPLE

Example 1: Temperature



Example 2: Is it Raining?



KEY POINTS

- The condition is tested. If True → Statement 1 runs.
- If False → Statement 2 runs.
- `if-else` helps the program choose one of two possible actions.

Conditional Statement: if-else Statement

- Complex conditional expressions are also possible

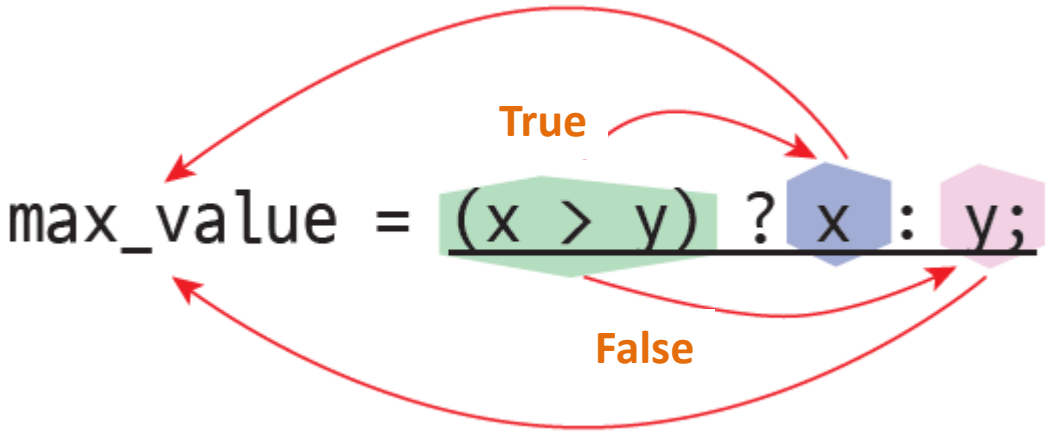
```
if ( score >= 80 && score < 90 )
    grade = 'B' ;
```

```
if ( ch == ' ' || ch == '\n' || ch == '\t' )
    white_space ++;
```

- Conditional **operator**: a simple if-else can be expressed with?

```
(score >= 60 ) ? printf( " Pass.\n" ) : printf( " Fail.\n" );
```

```
bonus = (( years > 30 ) ? 500 : 300 );
```



Coding Style

```
if (score >= 60)
      printf( "Passed\n" );
else
      printf( “Failed\n" );
```

```
if (score >= 60) {
      printf( "Passed\n" );
} else {
      printf( “Failed\n" );
}
```

```
if (score >= 60)
{
      printf( "Passed\n" );
}
else
{
      printf( “Failed\n" );
}
```

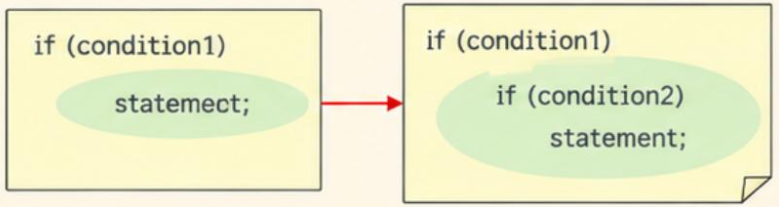
```
if (score >= 60) printf( "Passed\n" );
else  printf( “Failed\n" );
```

Conditional Statement: Nested if and the Matching Problem

- Nested if: an **if statement** contains another **if statement**.

1. Statement inside a statement

- You can place an **if** statement inside another sentence. (**nested if**)



2. else matches the nearest if

- An **else** clause is paired with the **closest** unmatched **if**.

Key Points

- A statement can contain another **if** statement.
- An **else** belongs to the nearest **if**.
- Use { } when you want an **else** to match a different **if** block.

Example 1: else matches the inner if

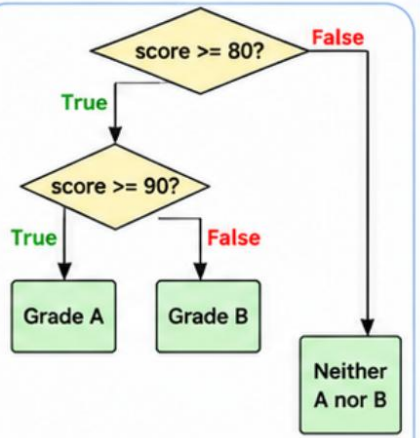
The **else** belongs to the nearest **if** (score >= 90).

```
1  if (score >= 80)
2      if (score >= 90)
3          printf("Your grade is A.\n");
4      else
5          printf("Your grade is B.\n");
```

Example 2: else matches the outer if

Braces { } change which **if** the **else** matches.

```
1  if (score >= 80) {
2      if (score >= 90)
3          printf("Your grade is A.\n");
4      }
5  else
6      printf("Your grade is neither an A nor a B.\n");
```



Conditional Statement: Consecutive if (if – else if – else)

1. SYNTAX

Used when we need to check multiple conditions in order.

GENERAL FORM

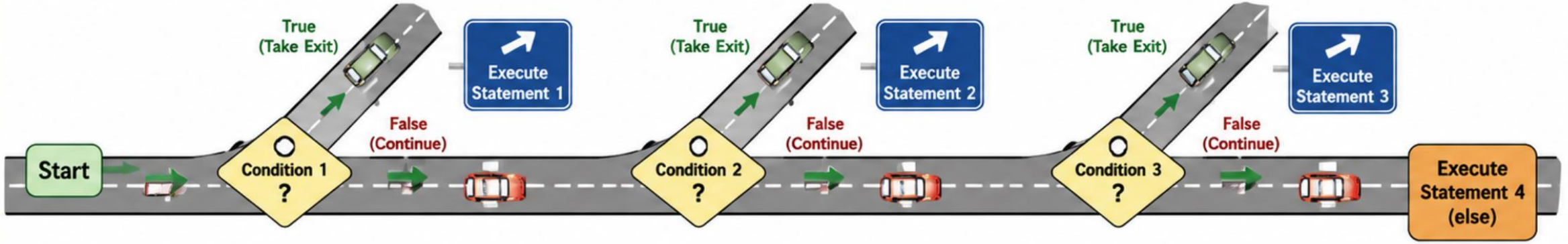
```
if ( condition1 )          statement 1;
else if ( condition2 )     statement 2;
else if ( condition3 )     statement 3;
else                       statement 4;
```

HOW IT WORKS

1. If condition 1 is true, statement 1 is executed.
2. Otherwise, if condition 2 is true, statement 2 is executed.
3. Otherwise, if condition 3 is true, statement 3 is executed.
4. Otherwise, statement 4 is executed.

2. FLOW (ROAD MAP)

The program checks conditions from top to bottom.



KEY POINTS

- Conditions are checked in order from top to bottom.
- The first true condition is executed, then the rest are skipped.
- If all conditions are false, the else block is executed.

Example: Determining Grades (if-else Chain)

```c
#include < stdio.h >

int main( void )
{
    int score;
    printf ( "Enter your grade: " );
    scanf ( "%d" , &score);

    if (score >= 90)      printf ( "Passed: Grade A\n" );
    else if (score >= 80) printf ( "Passed: Grade B\n" );
    else if (score >= 70) printf ( "Passed: Grade C\n" );
    else if (score >= 60) printf ( "Passed: Grade D\n" );
    else                  printf ( "Failed: Grade F\n" );
    return 0;
}
```

Example: Character Classification

```c
// Program to classify characters
#include < stdio.h >
int main( void )
{
    char ch ;
    printf ( "Enter a character: " );
    ch = getchar ();

    if ( ch >= 'A' && ch <= 'Z' )
         printf ( "%c is uppercase.\n" , ch );
    else if ( ch >= 'a' && ch <= 'z' )
         printf ( "%c is a lowercase letter.\n" , ch );
    else if ( ch >= '0' && ch <= '9' )
         printf ( "%c is a number.\n" , ch );
    else
         printf ( "%c is a miscellaneous character.\n" , ch );
    return 0;
}
```

switch Statement

Use switch when one variable can have many possible values.

1. SYNTAX

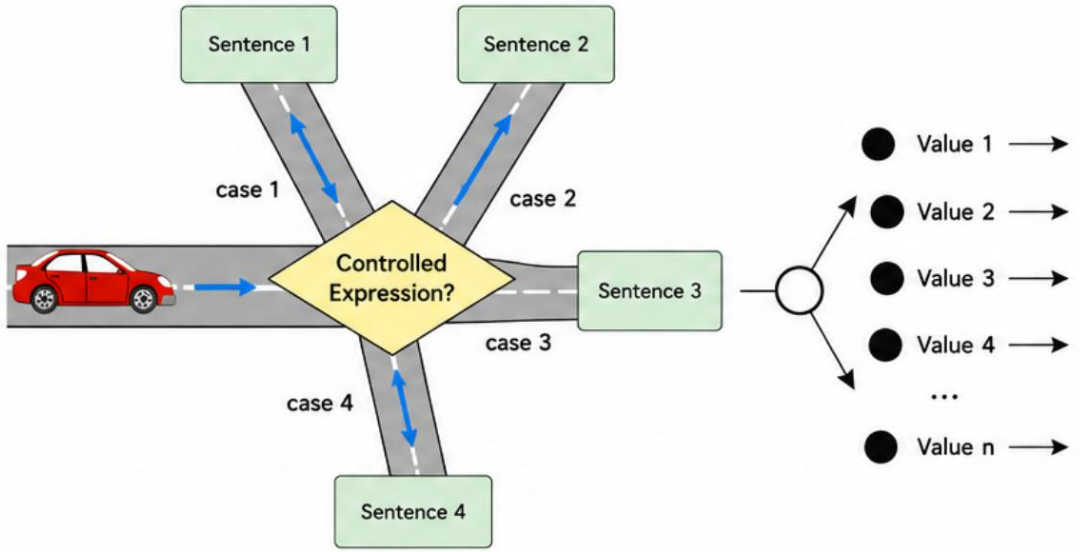
```
switch (controlled_expression)
{
    case value1:
        statement 1;
        break;
    case value2:
        statement 2;
        break;
        ...
    default:
        statement d;
        break;
}
```

If the value of the control expression is value1, statement 1 is executed.

If the value of the control expression is value2, statement 2 is executed.

If no matching value is found, this is executed.

2. FLOW (ROAD MAP)



3. HOW IT WORKS

1. The controlled expression is evaluated.
2. Its value is compared with each case value.
3. If a match is found, that case's statements are executed until break;.
4. If no match is found, the default case (if present) is executed.

4. EXAMPLE

```
int day = 3;
switch (day)
{
    case 1: printf("Mon\n"); break;
    case 2: printf("Tue\n"); break;
    case 3: printf("Wed\n"); break;
    default: printf("Invalid day\n");
}
```

If day = 3 → Output: Wed

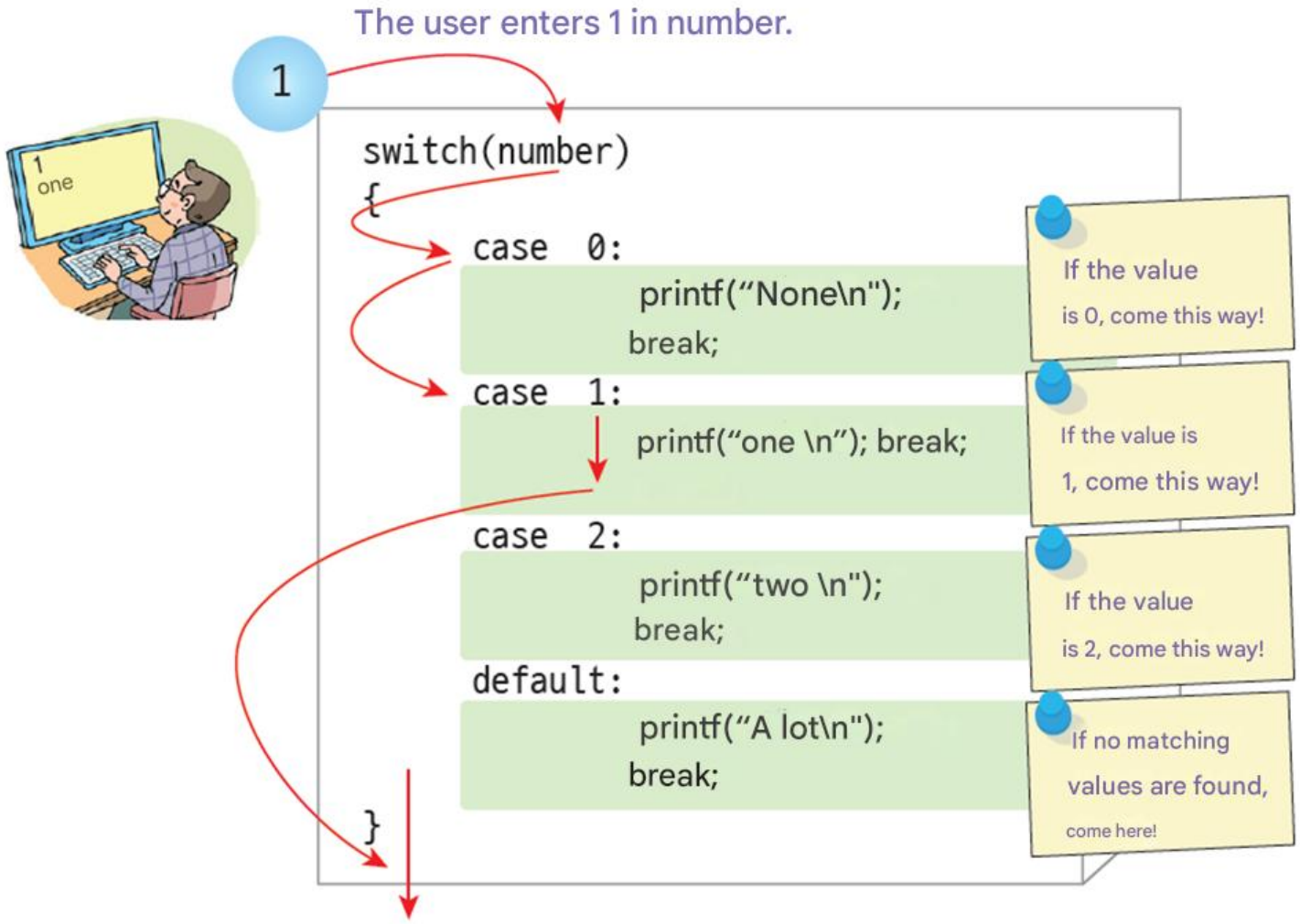
If day = 5 → Output: Invalid day

KEY TAKEAWAYS

- switch is ideal for multiple choices based on one variable.
- Use break; to stop execution after a case is matched.
- Always include default to handle unexpected values.

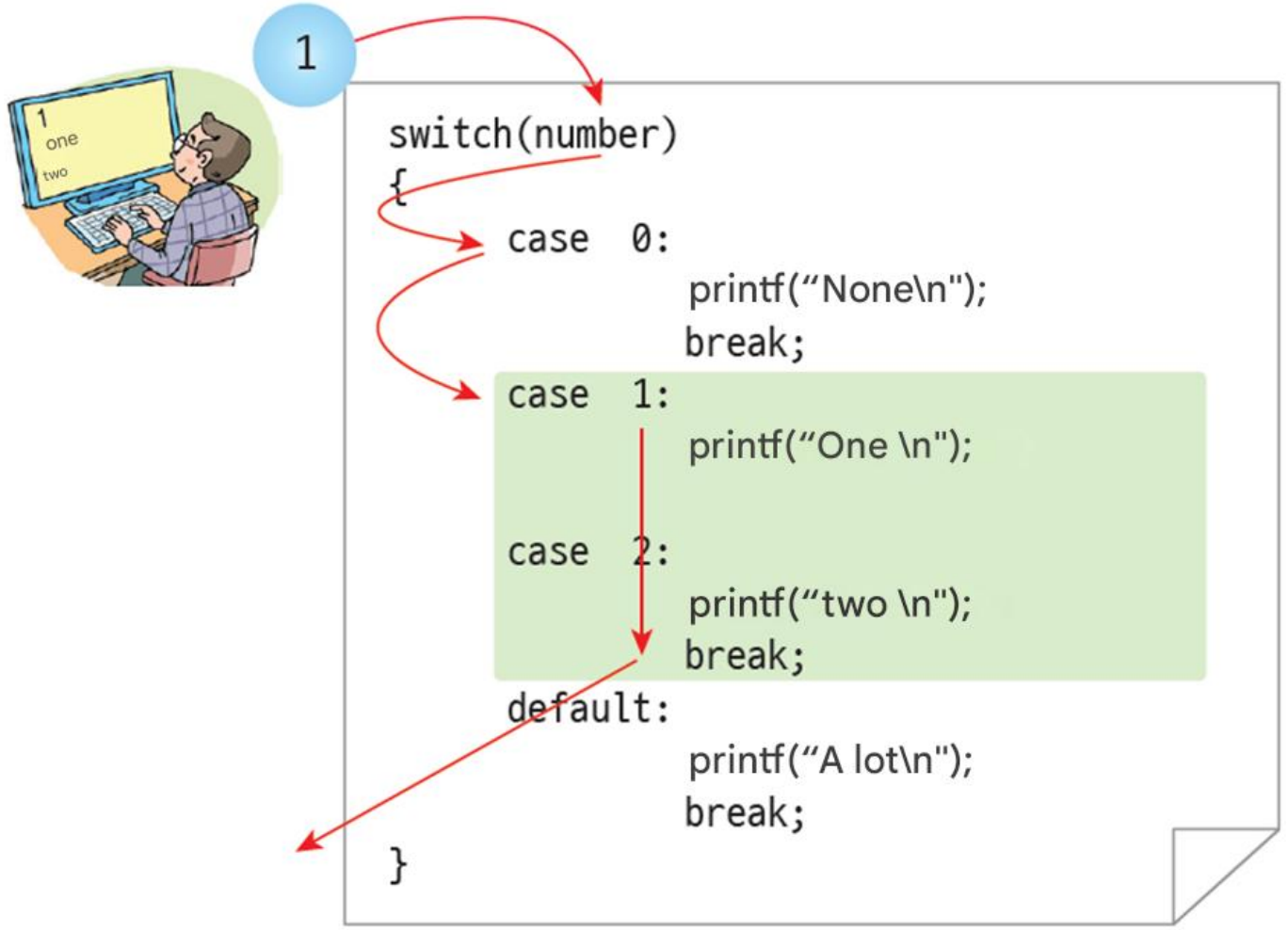
switch Statement: Case study 1

- User enter “1”



switch Statement: Case study 2

- If break is omitted



switch Statement: Case study 3

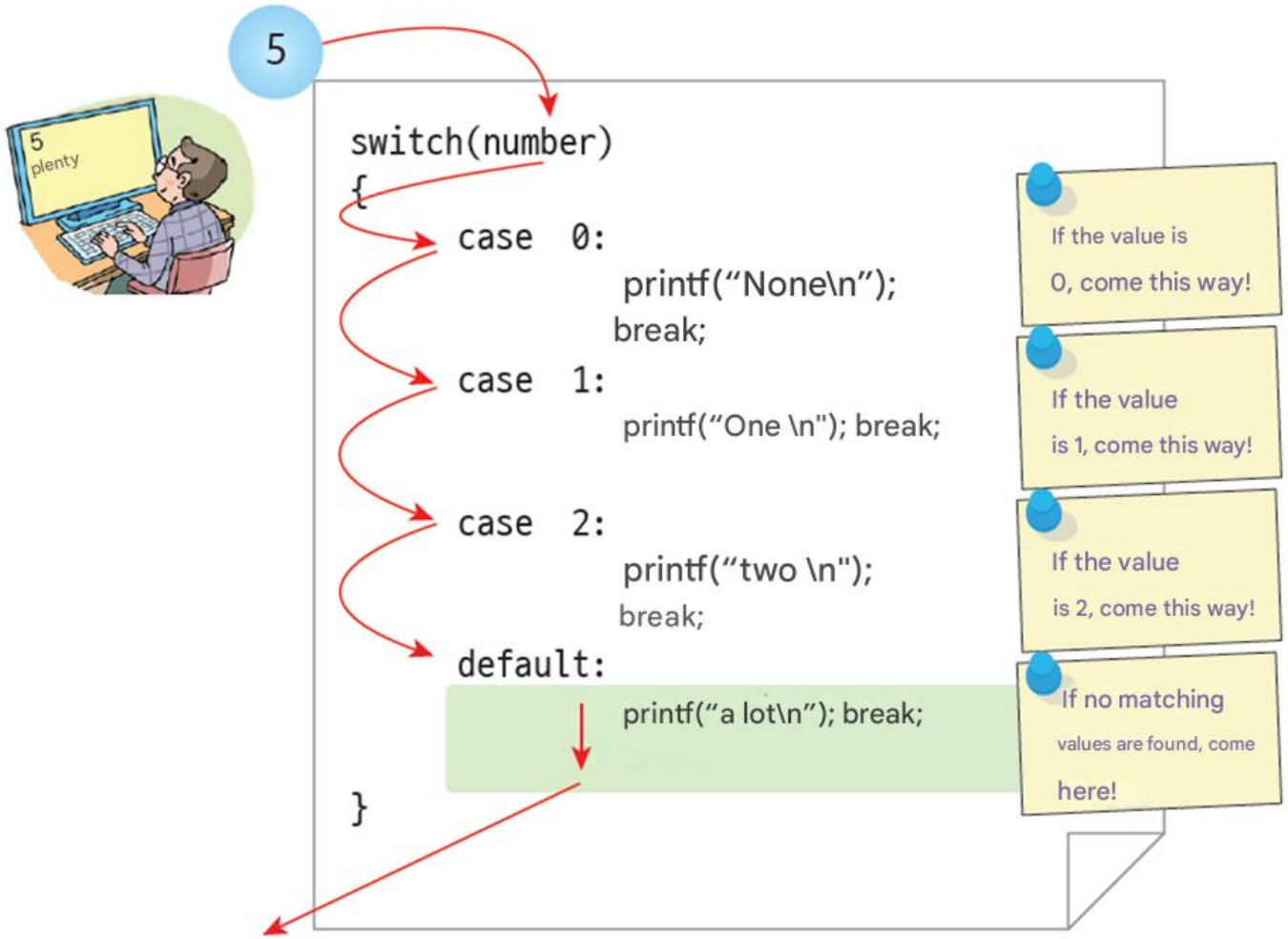
- Intentional omission of break: What is the intention?

2

```
switch(number)
{
    case  0:
            printf("None\n");
            break;
    case  1:
            printf("one \n");
            break;
    case  2:
    case  3:
            printf("two or three \n");
            break;
    default:
            printf("a lot\n");
            break;
}
```

switch Statement: Default

- User enter “5”



switch Statement: case label rule

Example

```
switch (number)
{
    case x:          // Variable cannot be used
            printf ( " Matches x . \n" );
            break ;
    case (x+2):      // Formulas containing variables
                     // cannot be used
            printf ( "Matches formula . \n" );
            break ;
    case 0.001:      // Real numbers cannot be used
            printf ( "error \n" );
            break ;
    case 'a' :       // OK! The character is can be used
            printf ( "character \n" );
            break ;
    case "abc":      // String cannot be used
            printf ( "string \n" );
            break ;
}
```

What Can and Cannot Be Used in case Labels?

Case Label	Example	Allowed?	Reason
Variable	`case x:`	✗ NO	Variables are not constant.
Expression with Variables	`case (x+2):`	✗ NO	Formulas containing variables are not allowed.
Real Number	`case 0.001:`	✗ NO	Real numbers (float, double) are not allowed.
Character (Constant)	`case 'a':`	✓ YES	Character constants are allowed.
String	`case "abc":`	✗ NO	Strings are not allowed.

IMPORTANT RULE

A case label must be a **CONSTANT INTEGER** OR **CONSTANT CHARACTER**.

(e.g., 1, 2, -5, 100, 'A', 'z', '\n')

QUICK SUMMARY

- Use integers (constants) and characters in case labels.
- Do not use variables, formulas with variables, real numbers, or strings.
- Always use `break;` to prevent fall-through.

switch Statement vs. if-else Statement

- A switch statement can always be rewritten using if-else.
- An if-else statement can be rewritten using switch only when the conditions compare one expression with constant values.

1. SWITCH → IF-ELSE (Always Possible)

SWITCH

```
switch (choice) {
  case 1:
     printf("Add\n");
     break;
  case 2:
     printf("Delete\n");
     break;
  case 3:
     printf("Exit\n");
     break;
  default:
     printf("Invalid choice\n");
}
```

EQUIVALENT IF-ELSE

```
if (choice == 1)
   printf("Add\n");
else if (choice == 2)
   printf("Delete\n");
else if (choice == 3)
   printf("Exit\n");
else
   printf("Invalid choice\n");
```

Each case in the switch becomes a condition in the if-else.

2. IF-ELSE → SWITCH (Only in Certain Cases)

IF-ELSE

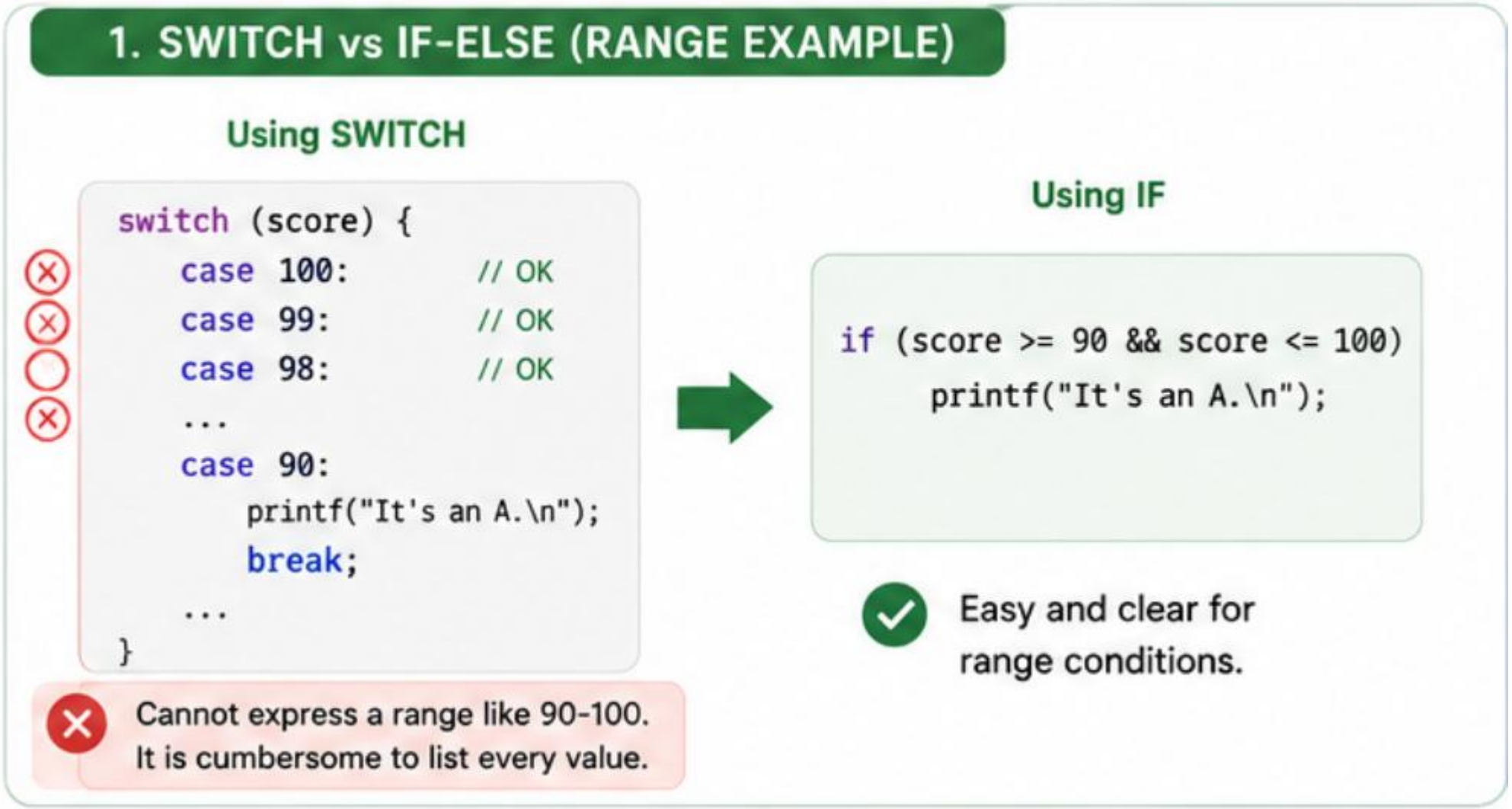
```
if (choice == 1)
   printf("Add\n");
else if (choice == 2)
   printf("Delete\n");
else if (choice == 3)
   printf("Exit\n");
else
   printf("Invalid choice\n");
```

EQUIVALENT SWITCH

```
switch (choice) {
  case 1:
     printf("Add\n");
     break;
  case 2:
     printf("Delete\n");
     break;
  case 3:
     printf("Exit\n");
     break;
  default:
     printf("Invalid choice\n");
}
```

Works when all conditions compare the same expression (choice) with different constant values (1, 2, 3, ...).

switch Statement vs. if-else Statement



switch Statement vs. if-else Statement

2. SWITCH EXAMPLE (GRADING SYSTEM)

```
int iscore;
...
iscore = score/ 10;      // Remainder is lost in integer division

switch (iscore) {
   case 9: grade = 'A'; break;      // 90-100 is A
   case 8: grade = 'B'; break;      // 80-89 is B
   case 7: grade = 'C'; break;      // 70-79 is C
   case 6: grade = 'D'; break;      // 60-69 is D
   default: grade = 'F'; break;     // 59 or less is F
}
```

How it works

1. The expression (iscore) is evaluated.
2. Its value is compared with each case label.
3. If a match is found, that case is executed.
4. If no match is found, `default` is executed.

goto statement

goto transfers control to a labeled statement within the same function.

1. HOW IT WORKS

Forward looking

```
goto label;
...
...
...
...
label:
Sentence #1;
Sentence #2;
Sentence #3;
...
```

Retrospective reference

```
label:
Sentence #1;
Sentence #2;
Sentence #3;
...
goto label;
```

A label is an identifier followed by a colon (:).
It marks the place to which control can be transferred.

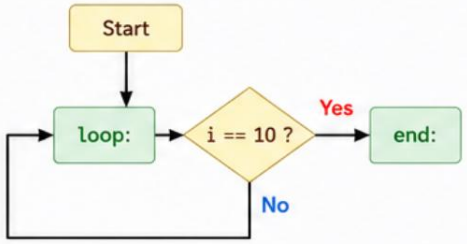
2. EXAMPLE PROGRAM

```c
// Multiplication table output program
#include <stdio.h>

int main(void)
{
    int i = 1;

loop:
    printf("%d * %d = %d \n", 3, i, 3 * i);
    i++;
    if (i == 10) goto end;
    goto loop;

end:
    return 0;
}
```

3. OUTPUT

```
3 * 1 = 3
3 * 2 = 6
3 * 3 = 9
3 * 4 = 12
3 * 5 = 15
3 * 6 = 18
3 * 7 = 21
3 * 8 = 24
3 * 9 = 27
```

Flow of control



4. NOTES

- Use goto carefully. It can make code hard to read.
- It is still used in some situations (error handling, cleanup, breaking out of nested loops, etc.).
- Do not jump outside the current function.

5. KEY POINTS

- goto jumps to a label in the same function.
- Labels mark target statements.
- Control can move forward or backward.
- Use it only when it improves program clarity.

Lab: Arithmetic Calculator (if-else && switch)

This program reads two integers and an operator (+, -, *, /),
performs the calculation, and prints the result.

1. Using if-else

```c
#include <stdio.h>

int main(void)
{
    int num1, num2;
    char op;

    printf("Enter two numbers: ");
    scanf("%d %d", &num1, &num2);

    printf("Enter operator (+, -, *, /): ");
    scanf(" %c", &op);

    if (op == '+')
        printf("Result = %d\n", num1 + num2);
    else if (op == '-')
        printf("Result = %d\n", num1 - num2);
    else if (op == '*')
        printf("Result = %d\n", num1 * num2);
    else if (op == '/')
        printf("Result = %d\n", num1 / num2);
    else
        printf("Invalid operator\n");

    return 0;
}
```

↔

2. Using switch

```c
#include <stdio.h>

int main(void)
{
    int num1, num2;
    char op;

    printf("Enter two numbers: ");
    scanf("%d %d", &num1, &num2);
    printf("Enter operator (+, -, *, /): ");
    scanf(" %c", &op);

    switch (op)
    {
        case '+':
            printf("Result = %d\n", num1 + num2);
            break;
        case '-':
            printf("Result = %d\n", num1 - num2);
            break;
        case '*':
            printf("Result = %d\n", num1 * num2);
            break;
        case '/':
            printf("Result = %d\n", num1 / num2);
            break;
        default:
            printf("Invalid operator\n");
            break;
    }

    return 0;
}
```

Q & A

Lab & Quiz: right after this - see you in the practice session!